

Practical week - 5

MySQL Workbench Code:

```
CREATE DATABASE library_db; USE
library_db;

CREATE TABLE IF NOT EXISTS members ( id INT
    AUTO_INCREMENT PRIMARY KEY,
    username VARCHAR(100) UNIQUE NOT NULL,
    password_hash VARCHAR(255) NOT NULL,
    role VARCHAR(30) NOT NULL DEFAULT 'member'
);

CREATE TABLE IF NOT EXISTS books (
    id INT AUTO_INCREMENT PRIMARY KEY, title
    VARCHAR(250) NOT NULL,
    author VARCHAR(200) NOT NULL,
    isbn VARCHAR(13) UNIQUE NOT NULL,
    price DECIMAL(10,2) NOT NULL,
    published_date DATE NOT NULL, description
    TEXT,
    embedding JSON
);
```

database.py

```
from sqlalchemy.ext.asyncio import (
    AsyncSession, async_sessionmaker, create_async_engine
)

from sqlalchemy.orm import DeclarativeBase DATABASE_URL = (
    "mysql+asyncmy://root:YOUR_MYSQL_PASSWORD" "@localhost:3306/library_db"
)

engine = create_async_engine(DATABASE_URL, echo=True)
AsyncSessionLocal = async_sessionmaker(
```

```

        engine, expire_on_commit=False
    )
class Base(DeclarativeBase): pass
async def get_db():
    async with AsyncSessionLocal() as session: yield
        session

```

models.py

```

from datetime import date from
decimal import Decimal

from sqlalchemy import Date, Integer, Numeric, String, Text, JSON from
sqlalchemy.orm import Mapped, mapped_column
from database import Base class
Member(Base):
    __tablename__ = "members"
    id: Mapped[int] = mapped_column( Integer,
        primary_key=True
    )
    username: Mapped[str] = mapped_column(
        String(100), unique=True, nullable=False
    )
    password_hash: Mapped[str] = mapped_column( String(255),
        nullable=False
    )
    role: Mapped[str] = mapped_column( String(30),
        default="member", nullable=False
    )
class Book(Base):
    __tablename__ = "books"
    id: Mapped[int] = mapped_column( Integer,
        primary_key=True
    )
    title: Mapped[str] = mapped_column(

```

```

        String(250), nullable=False
    )
    author: Mapped[str] = mapped_column(
        String(200), nullable=False
    )
    isbn: Mapped[str] = mapped_column( String(13),
        unique=True, nullable=False
    )
    price: Mapped[Decimal] = mapped_column(
        Numeric(10, 2), nullable=False
    )
    published_date: Mapped[date] = mapped_column( Date,
        nullable=False
    )
    description: Mapped[str | None] = mapped_column( Text,
        nullable=True
    )
    embedding: Mapped[list[float] | None] = mapped_column( JSON,
        nullable=True
    )

```

schemas.py

```

from datetime import date from
decimal import Decimal
from pydantic import BaseModel, Field, field_validator class
BookSchema(BaseModel):
    title: str
    author: str
    isbn: str = Field(
        min_length=13,
        max_length=13
    )
    price: Decimal = Field(gt=0)

```

```

published_date: date description: str |
None = None
@field_validator("published_date")
@classmethod
def validate_published_date(cls, value: date): if value
    > date.today():
        raise ValueError(
            "published_date cannot be in the future"
        )
    return value

class MemberRegister(BaseModel): username: str
    password: str = Field(min_length=6) class
TokenResponse(BaseModel):
    access_token: str
    token_type: str

```

security.py

```

from datetime import datetime, timedelta, timezone from fastapi
import Depends, HTTPException
from fastapi.security import OAuth2PasswordBearer from jose
import JWTErrors, jwt
from passlib.context import CryptContext from
sqlalchemy import select
from sqlalchemy.ext.asyncio import AsyncSession from
database import get_db
from models import Member SECRET_KEY =
"CHANGE_THIS_SECRET" ALGORITHM =
"HS256" ACCESS_TOKEN_EXPIRE_MINUTES = 60
pwd_context = CryptContext(
    schemes=["bcrypt"],
    deprecated="auto"
)
oauth2_scheme = OAuth2PasswordBearer(

```

```

        tokenUrl="/login"
    )
def hash_password(password: str): return
    pwd_context.hash(password)
def verify_password(
    plain_password,
    hashed_password
):
    return pwd_context.verify(
        plain_password, hashed_password
    )
def create_access_token(username, role):
    expire = datetime.now(timezone.utc) + timedelta(
        minutes=ACCESS_TOKEN_EXPIRE_MINUTES
    )
    payload = {
        "sub": username,
        "role": role, "exp":
        expire
    }
    return jwt.encode(
        payload, SECRET_KEY,
        algorithm=ALGORITHM
    )
async def get_current_member(
    token: str = Depends(oauth2_scheme), db:
    AsyncSession = Depends(get_db)
):
    credentials_exception = HTTPException(
        status_code=401,
        detail="Invalid or expired token", headers={
            "WWW-Authenticate": "Bearer"
        }
    )

```

```

)
try:
    payload = jwt.decode(
        token,
        SECRET_KEY,
        algorithms=[ALGORITHM]
    )
    username = payload.get("sub") if not
    username:
        raise credentials_exception
except JWTError:
    raise credentials_exception
result = await db.execute(
    select(Member).where( Member.username
        == username
    )
)
member = result.scalar_one_or_none() if not
member:
    raise credentials_exception
return member

```

embedding.py

```

import hashlib import
math VECTOR_DIM =
384
def text_to_vector(text: str) -> list[float]: values = []
    for i in range(VECTOR_DIM): digest =
        hashlib.sha256(
            f"{text.lower()}:{i}".encode()
        ).digest()
        number = int.from_bytes(
            digest[:4], byteorder="big"
        )

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```

        values.append(
            (number / 2**32) * 2 - 1
        )
    magnitude = math.sqrt( sum(x * x
        for x in values)
    )
    return [
        x / magnitude
        for x in values
    ]

```

main.py

```

from fastapi import (
    BackgroundTasks,
    Depends,
    FastAPI,
    HTTPException,
    Query,
    status
)
from fastapi.security import OAuth2PasswordRequestForm
from sqlalchemy import select
from sqlalchemy.ext.asyncio import AsyncSession
from database import Base, engine, get_db
from models import Book, Member
from schemas import BookSchema, MemberRegister, TokenResponse
from security import (
    create_access_token,
    get_current_member,
    hash_password, verify_password
)
from embedding import text_to_vector
import math

app = FastAPI(
    title="Library Backend & Security API",
    version="1.0"
)

# Create tables when application starts
@app.on_event("startup")
async def startup():
    async with engine.begin() as connection:
        await connection.run_sync(
            Base.metadata.create_all

```

```

    )
# Home @app.get("/")
async def home():
    return {
        "message": "Library API is running"
    }
# Background task for logging
def log_new_book(book_id: int, title: str): with
    open(
        "book_events.txt", "a",
        encoding="utf-8"
    ) as file:
        file.write(
            f"New Book Added: id={book_id}, "
            f"title={title}\n"
        )
# Cosine similarity def
cosine_similarity(
    a: list[float],
    b: list[float]
) -> float:
    if len(a) != len(b): return -
        1.0
    dot = sum( x
        * y
        for x, y in zip(a, b)
    )
    norm_a = math.sqrt( sum(x *
        x for x in a)
    )
    norm_b = math.sqrt( sum(y *
        y for y in b)
    )
    if norm_a == 0 or norm_b == 0: return 0.0
    return dot / (norm_a * norm_b) #
Register member @app.post("/register")
async def register(
    member: MemberRegister,
    db: AsyncSession = Depends(get_db)
):
    result = await db.execute(
        select(Member).where(
            Member.username == member.username
        )
    )
    if result.scalar_one_or_none(): raise
        HTTPException(
            status_code=400, detail="Username

```

```

        already exists"
    )
    new_member = Member(
        username=member.username,
        password_hash=hash_password(
            member.password
        ),
        role="member"
    )
    db.add(new_member) await
    db.commit()
    await db.refresh(new_member)
    return {
        "message": "Member registered successfully", "username":
        new_member.username,
        "role": new_member.role
    }
# Login
@app.post(
    "/login", response_model=TokenResponse
)
async def login(
    form_data: OAuth2PasswordRequestForm = Depends(), db:
    AsyncSession = Depends(get_db)
):
    result = await db.execute(
        select(Member).where(
            Member.username == form_data.username
        )
    )
    member = result.scalar_one_or_none() if (
        not member
        or not verify_password( form_data.password,
            member.password_hash
        )
    ):
        raise HTTPException(
            status_code=401,
            detail="Invalid username or password"
        )
    token = create_access_token(
        member.username, member.role
    )
    return {
        "access_token": token,
        "token_type": "bearer"
    }
# Add book - protected by JWT @app.post(

```

```

    "/books",
    status_code=status.HTTP_201_CREATED
)
async def create_book( book:
    BookSchema,
    background_tasks: BackgroundTasks, current_member:
    Member = Depends(
        get_current_member
    ),

    db: AsyncSession = Depends(get_db)
):
    # Create embedding embedding =
    text_to_vector(
        f"{book.title} "
        f"{book.author} "
        f"{book.description or ""}"
    )
    new_book = Book(
        title=book.title,
        author=book.author,
        isbn=book.isbn,
        price=book.price,
        published_date=book.published_date,
        description=book.description,
        embedding=embedding
    )
    db.add(new_book)
    await db.commit()
    await db.refresh(new_book)
    # Schedule background logging
    background_tasks.add_task(
        log_new_book, new_book.id,
        new_book.title
    )
    return {
        "message": "Book added successfully", "book_id":
        new_book.id,
        "added_by": current_member.username
    }
# Search books using cosine similarity
@app.get("/books/search")
async def search_books(
    q: str = Query(min_length=1), limit:
    int = Query(
        default=5,
        ge=1, le=20
    ),
    current_member: Member = Depends(

```

```

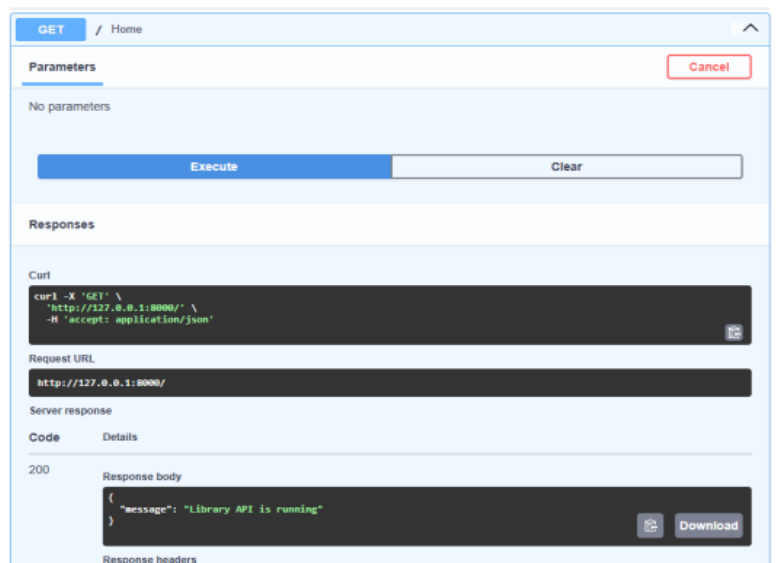
        get_current_member
    ),
    db: AsyncSession = Depends(get_db)
):
    query_vector = text_to_vector(q) result =
    await db.execute(
        select(Book).where(
            Book.embedding.is_not(None)
        )
    )

    books = result.scalars().all() ranked =
    []
    for book in books:
        if book.embedding:

            similarity = cosine_similarity(
                query_vector, book.embedding
            )
            ranked.append({ "id":
                book.id, "title":
                book.title,
                "author": book.author,
                "isbn": book.isbn, "price":
                float(book.price),
                "published_date":
                book.published_date, "description":
                book.description,
                "similarity":
                similarity
            })
    ranked.sort(
        key=lambda item: item["similarity"],
        reverse=True
    )
    return {
        "query": q,
        "results": ranked[:limit]
    }

```

GET:



1.

2. Register

POST/register



After execution:

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/register' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "username": "Chaitanya",
    "password": "chaitanya"
  }'
```

Request URL

http://127.0.0.1:8000/register

Server response

Code	Details
200	Response body <pre>{ "message": "Member registered successfully", "username": "Chaitanya", "role": "member" }</pre> Response headers <pre>content-length: 86 content-type: application/json date: Sat, 13 Sep 2025 05:26:35 GMT server: uvicorn</pre>

Authorization:

Available authorizations



Scopes are used to grant an application different levels of access to data on behalf of the end user. Each API may declare one or more scopes.

API requires the following scopes. Select which ones you want to grant to Swagger UI.

OAuth2PasswordBearer (OAuth2, password)

Authorized

Token URL: /login

Flow: password

username: Chaitanya2026

password: *****

Client credentials location: basic

client_secret: *****

Logout

Close

3.

4. Login

POST /login

POST

/login Login

Parameters

No parameters

Request body required

grant_type

string | (string | null)
pattern: "password\$"

password

☐ Send empty value

username * required

string

Chaitanya2026

☐ Send empty value

password * required

string(\$password)

☐ Send empty value

scope

string

scope

☒ Send empty value

client_id

string | (string | null)

string

☐ Send empty value

client_secret

string | (string | null)(\$password)

☐ Send empty value

Response:

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/login' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/x-www-form-urlencoded' \
  -d 'grant_type=password&username=Charan2026&password=priyanka123&scope=&client_id=string&client_secret=*****'
```

Request URL

http://127.0.0.1:8000/login

Server response

Code	Details
200	Response body <pre>{ "access_token": "eyJhdGUiOiJ1ZjI1NiIsIn85Ci6IkpXVCJ9.eyJ2ZWZlIjQcm15bmFrcyYmYjIiLCJyb2x1IjoibWVYmVYiIiwia2NwIjoxOjtzg5MTk0NjUyYzQ-VyjbXurfr4QcLbngdJIPa4I-g4vdbI75bCJBaoU3jmu0", "token_type": "bearer" }</pre> Response headers <pre>content-length: 196 content-type: application/json date: Sat, 12 Sep 2026 05:30:51 GMT server: uvicorn</pre>

5. POST /books

POST

/books

Create Book

Parameters

Cancel

Reset

No parameters

Request body

required

application/json

Edit Value

Schema

```
{
  "title": "Chaitanya's Guide to Java",
  "author": "Chaitanya",
  "isbn": "9781234567897",
  "price": 599.00,
  "published_date": "2025-08-15",
  "description": "A beginner friendly guide to Java programming and data structures."
}
```

Response:

Responses

Curl

```
curl -X 'POST' \
  "http://127.0.0.1:8000/books" \
  -H 'accept: application/json' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVC9i.eyJhdGU0Ij0QXo1SM6rTTIuMTY1LQyNzJj0l0MTWZyIueI7SOx5jonkcg8ATeMWT781Q?RBAU7O@wPw4tIk53no2977Maf11xwLoet6r-f65Q318' \
  -H 'Content-Type: application/json' \
  -d '{
    "title": "Chaitanya\'s Guide to Java",
    "author": "Chaitanya",
    "isbn": "9781234567897",
    "price": 599.00,
    "published date": "2025-08-15",
    "description": "A beginner friendly guide to Java programming and data structures."
  }'
```

Request URL

http://127.0.0.1:8000/books

Server response

Code	Details
201	Response body <pre>{ "message": "Book added successfully", "book_id": 1, "added_by": "Chaitanya2026" }</pre> Response headers <pre>content-length: 75 content-type: application/json date: Sat, 13 Sep 2025 05:31:53 GMT server: uvicorn</pre>

6. GET /books/search

GET

/books/search Search Books

🔒 🔑

Parameters

Name

Description

q * required

string

(query)

Java

minLength: 1

limit

integer

(query)

5

maximum: 20
minimum: 1

Execute

Clear

Response:

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/books/search?q=Java&limit=5' \
  -H 'accept: application/json' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ2aW0iOiJ0c15mfjYTMWYjI1CiJyY2x1IjoibW9tYmVyIiwicm96IjoibG92PTk0N9j=FQ_VyJ8xUrf4QCLbfGwUIP4t-q4v6b1750CJ8aoU3jme0'
```

Request URL

```
http://127.0.0.1:8000/books/search?q=Java&limit=5
```

Server response

Code	Details
200	Response body <pre>{ "query": "Java", "results": [{ "id": 1, "title": "Chaitanya's Guide to Java", "author": "Chaitanya", "isbn": "9781234567897", "price": 599.00, "published_date": "2025-08-15", "description": "A beginner friendly guide to Java programming and data structures.", "similarity": 0.888868293218186619 }, { "id": 2, "title": "Python for Beginners", "author": "John Smith", "isbn": "9781234567898", "price": 499.00, "published_date": "2024-06-10", "description": "An introduction to Python programming.", "similarity": 0.822237869296884439 }] }</pre> Response headers <pre>content-length: 494 content-type: application/json date: Sat, 12 Sep 2026 05:31:04 GMT server: uvicorn</pre>

7. Test validation

Test A — Invalid ISBN

POST /books Create Book

Parameters

No parameters

Request body required application/json

Edit Value | Schema

```
{
  "title": "Test Book",
  "author": "Test Author",
  "isbn": "12345",
  "price": 100,
  "published_date": "2024-01-01",
  "description": "Test"
}
```

Response will be 422 error

Server response

Code	Details
422	Error: Unprocessable Entity Response body <pre>{ "detail": [{ "type": "string_too_short", "loc": ["body", "isbn"], "msg": "String should have at least 13 characters", "input": "12345", "ctx": { "min_length": 13 } }] }</pre>

Test B — Negative price

POST /books Create Book

Parameters

Cancel

Reset

No parameters

Request body required

application/json

Edit Value

Schema

```
{
  "title": "Test Book",
  "author": "Test Author",
  "isbn": "1234567890100",
  "price": -100,
  "published_date": "2024-01-01",
  "description": "Test"
}
```

Response will be 422 because price<0

Code

Details

422

Error: Unprocessable Entity

Response body

```
{
  "detail": [
    {
      "type": "greater_than",
      "loc": [
        "body",
        "price"
      ],
      "msg": "Input should be greater than 0",
      "input": -100,
      "ctx": {
        "gt": 0
      }
    }
  ]
}
```

Download

Test C — Future publication date

POST /books Create Book

Parameters

Cancel

Reset

No parameters

Request body required

application/json

Edit Value

Schema

```
{
  "title": "Future Book",
  "author": "Test Author",
  "isbn": "1234567890124",
  "price": 100,
  "published_date": "2035-01-01",
  "description": "Test"
}
```

Response will be 422 error because a book cannot have a future publication date

Server response	
Code	Details
422	Error: Unprocessable Entity Response body <pre>{ "detail": [{ "type": "value_error", "loc": ["body", "published_date"], "msg": "Value error, published_date cannot be in the future", "input": "2035-01-01", "ctx": { "error": {} } }] }</pre>  Download